

Status-Based Influence Processes: The Role of Norm Salience in Contagion of Adolescent Risk Attitudes

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This study examined whether classroom norms based on status explained between-class variations in selection processes and particularly influence processes on adolescents' risk attitudes in a sample of 1092 adolescents (age 12–13) across 47 classrooms. Based on the association between status (popularity) and risk attitudes (*norm salience*), it was hypothesized that risk attitudes would proliferate more via peer influence processes in classes with a more positive correlation between status and risk attitudes, compared with classes with a somewhat positive or neutral correlation. Results were in line with these expectations. These findings show that classroom norms based on status affect adolescents' susceptibility to peer influence on risk attitudes in friendship networks, suggesting status-based influence processes.

Peer relations and friendships are important for the development of behaviors and attitudes during adolescence (Dishion, Piehler, & Myers, 2008). Friendships with classmates become more important to adolescents as they increasingly interact with each other (Newcomb & Bagwell, 1995). Particularly with regard to risk behaviors, such as alcohol use (Burk, Van der Vorst, Kerr, & Stattin, 2012), smoking (Mercken, Snijders, Steglich, & De Vries, 2009), and delinquency (De Cuyper, Weerman, & Rutter, 2009), peers are known to be influential upon adolescents' behavior (see for an overview Veenstra & Dijkstra, 2011).

Remarkably, studies into the effects of selection and influence on risk behavior among adolescents consistently show considerable variation between school classes, grades, or schools in the extent to which selection or influence takes place (cf. Veenstra & Steglich, 2012). Findings from some studies show that friends are selected on the basis of similar risk behavior, such as delinquency (Knecht, Snijders, Baerveldt, Steglich, & Raub, 2010), smoking (Mercken, Snijders, Steglich, Vartiainen, & De Vries, 2010), and alcohol use (Knecht, Burk, Weesie, & Steglich, 2011), whereas these dimensions

are less or not relevant in other studies (e.g., delinquency: Weerman, 2011; smoking: Pearson, Steglich, & Snijders, 2006; alcohol use: Mercken et al., 2009).

Similarly, friends influence each other's risk behavior in some classes (delinquency: Weerman, 2011) but not in others (delinquency: Light & Dishion, 2007; smoking: Mercken et al., 2009). Both selection and influence processes also occur in classes at the same time (delinquency: De Cuyper et al., 2009; smoking: Mercken et al., 2010; alcohol use: Knecht et al., 2011). So far, studies have only reported this variation; an explanation of why these processes of selection and influence differ between classes is lacking.

In this study, we aim to understand why classes differ in selection but particularly in influence processes by examining both processes simultaneously in a large sample of adolescents across different classrooms. We propose that norms about risk behavior within a classroom as contextual factors may explain between-class variations. Specifically, we consider the attitudes of high-status (popular) adolescents as potentially important norms that might explain why in some classes influence processes in risk attitudes are more likely than in others (cf. Henry et al., 2000).

Whereas previous research mainly addressed risk behaviors, we focus in our study on attitudes toward risk behaviors (risk attitudes). Although there is a known discrepancy between attitudes and actual behaviors (Festinger, 1957; Funder &

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Colvin, 1991; Megens & Weerman, 2010), recent research has found risk attitudes as a precursor for adolescents' actual risk behaviors (Greene & Banerjee, 2009; Snyder & Fleming-Milici, 2009). Moreover, similarly to risk behavior, friends tend to be more similar in attitudes toward risk behaviors (Fang, Stanton, Li, Galbraith, & Feigelman, 1996).

Background

Basically, norms can be considered as injunctive, reflecting normative beliefs about what is appropriate, accepted or not, or descriptive, reflecting the average behavior of all peers (Deutsch & Gerard, 1955). Particularly, this latter approach has been applied in peer relations research, showing that descriptive norms matter for how behaviors are evaluated in the peer group. Starting with the work of Wright, Giammarino, and Parad (1986), who showed that aggressive behavior was only detrimental to peer acceptance in low aggressive groups and not in high aggressive groups, other studies replicated (Boivin, Dodge, & Coie, 1995; Chang, 2004; Stormshak, Bierman, Bruschi, Dodge, & Coie, 1999) and extended this line of research by looking at positive interactions (Boivin et al., 1995), prosociality (Chang, 2004), social withdrawal (Chang, 2004; Stormshak et al., 1999), and bullying (Sentse, Scholte, Salmivalli, & Voeten, 2007). Together, these studies demonstrated that deviation from the average behavior of peers puts children at risk for lower acceptance or even rejection in the peer group.

Because being accepted by peers is important to gain a certain sense of belonging (Tarrant et al., 2001), and affection and trust (Leenders, 1996), adolescents are susceptible to conforming to group norms in order to fit in. Importantly, pressures from the outside (i.e., norms) can give rise to behave in ways that are expected in the group (Ajzen, 1991). This suggests that the behavior or attitudes of peers influence adolescents' own behaviors and attitudes and that adolescents are influenced to conform to what is considered as normative and appropriate.

Yet, it is unlikely that all peers are equally influential. Peer relations are organized along the dimension of status (popularity), reflecting a hierarchical ordering in the group. It is commonly assumed that popular adolescents are attractive for affiliation (Dijkstra, Cillessen, & Borch, in press), more powerful (Adler & Adler, 1995), visible (Lease, Kennedy, & Axelrod, 2002), dominant (Parkhurst & Hopmeyer, 1998), attractive (LaFontana & Cillessen, 2002), and influential than others (Adler, Kless, & Adler, 1992).

In adolescence, the achievement of status in the peer groups becomes more important (Corsaro & Eder, 1990; Ojanen, Gronroos, & Salmivalli, 2005). Because adolescents generally strive for status, the behavior of popular peers is seen in a more positive light (see Dijkstra, Lindenberg, Verhulst, Ormel, & Veenstra, 2009). Conforming to the behaviors of popular peers might increase the own popularity directly or via affiliation with high-status peers (Dijkstra, Cillessen, Lindenberg, & Veenstra, 2010). For that reason, social influence might be driven more by the behaviors and attitudes of popular adolescents than by those of all peers. Hence, popular adolescents are likely to set the norm among peers (Dijkstra, Lindenberg, & Veenstra, 2008). One approach to capture the norms of popular adolescents is by looking at the within-classroom correlation between status and behaviors or attitudes, the so-called *norm salience* (Henry et al., 2000).

In this study, we apply this norm salience approach to attitudes of popular adolescents, reflecting injunctive norms. If popular adolescents are truly more influential than others are, influence processes regarding attitudes toward risk behaviors might be stronger in classes where the more high-status adolescents have more positive attitudes toward risk behaviors (i.e., the more status correlates positively with risk attitudes). Thus, we expected that risk attitudes would proliferate more via peer influence processes in classes where risk attitudes are more positively associated with status than in classes where risk attitudes are unrelated or less positively associated with status.

To rule out the possibility that norm salience accounts for differences in influence processes between classrooms rather than the average attitudes of all classmates, we alternatively tested to what extent the average of risk attitudes in classes explains between-classroom variation in influence processes.

The Present Study

The present study examined whether the extent to which peer influence on risk attitudes proliferates differently across contexts depends on the injunctive norm salience in a given context. Peer influence on risk attitudes and friendship selection processes were compared across three types of classes in which norm salience differed at the beginning of the school year. Based on the correlation of popularity (measuring status) and risk attitudes shortly after students had entered a new

school, we differentiated between classes with a neutral, a somewhat positive, and a more positive association between status and risk attitudes. We then investigated whether the development of risk attitudes and friendships were determined by different processes in these classrooms.

Instead of modeling complex interactions to capture the influence of popular students directly, our injunctive norm salience approach captures both direct and indirect influence processes. Whereas the direct influence of popular students pertains to changes in risk attitudes among popular students' friends, the indirect influence process covers the extent to which adolescents who are not friends with popular students conform to the risk attitudes of popular students through observation rather than through direct contact.

Using longitudinal social network analysis, we were able to overcome major shortcomings of previous research on influence and selection processes, that is, untangle both processes simultaneously, while taking into account the role of structural network effects and dependencies in the data by looking at the complete networks (cf. Steglich, Snijders, & Pearson, 2010). We differentiated between school classes (not schools) because the students in our sample spent most of their time in their own classroom, resulting in more interaction between peers and potentially more status-based influence. Hence, the classroom is a relevant context to study development (Stark & Flache, 2012). To our knowledge, this study is the first to explain between-class variations in the effects of selection and influence on adolescents' risk attitudes looking at the role of classroom context by means of norm salience.

METHOD

Sample

Data were drawn from The Arnhem School Study (TASS), a three-wave longitudinal study among 1350 students from the first year of secondary education (age 12–13) in the city of Arnhem, a mid-sized city in the Netherlands (Stark & Flache, 2012). Data were collected 2 and 3 weeks after students transferred to secondary education (September 2008), 3 months later (December 2008), and six additional months later (June 2009). Sixty-one of all the first-year secondary school classes in the city of Arnhem (88.4 percent) participated in the study. Of the 1350 students in these classes, 1225 actually participated (response rate = 90.7%). Seven classes did not participate in all three waves and were

excluded from the sample. We further excluded seven additional classes ($n = 133$) because of estimation problems (i.e., did not satisfy the model convergence requirements; see Snijders, Steglich, & Schweinberger, 2007). Logistic regression at the classroom level identified that the number of students was a significant predictor of model convergence ($OR = 1.15$, $p < .001$). Because this predictor is partially related to the power of the analyses (amount of available information), we suspect that nonconvergence was mainly the result of low power.

There were 1092 students in the remaining 47 classes, of whom 1006 (92.1%) participated in Wave 1, 999 (91.5%) in Wave 2, and 976 (89.4%) in Wave 3. Missing data on risk attitudes due to nonparticipation was missing at random ($X^2 [9] = 14.94$, $p = .09$; Little's MCAR test). Risk attitudes did not differ between students with missing data and students with complete data (Wave 1: $t[994] = 0.77$, $p = 0.44$; Wave 2: $t[1050] = 1.05$, $p = 0.29$; Wave 3: $t[955] = -.38$, $p = 0.70$). There were on average 22.6 ($SD = 5.45$) students in each class at Wave 1. The final sample consisted of 563 boys, 502 girls, and 27 students who did not participate in any wave, leaving their sex unknown. These students were also included because they could be nominated as best friends by classmates, making them part of a friendship network.

Before the first wave, parents received a letter informing them about the study and offering them the opportunity to decline to have their child participate. Furthermore, students were informed that their answers would be treated confidentially and that their participation was entirely voluntary. All students from a class who were present on the day of the data collection simultaneously completed the questionnaires online on separate computers in a school computer laboratory. A teacher read the instructions to the students and supervised the completion of the questionnaires, which took on average 30 minutes per wave. Some students who were absent at Wave 1 filled in the questionnaire within one month after the regular assessment (0.46%).

Measures

Friendship Networks were measured by asking participants to nominate the classmates whom they perceived as "best friends." A list of all classmates (excluding their own name) was provided on the computer screen at all observations. Participants could nominate friends by marking the box next to

the name. Friendship nominations were coded 1; non-nominations were coded 0. Participants could provide unlimited same- or cross-sex nominations. Respondents nominated on average 3.99 ($SD = 3.21$) classmates as best friends at Wave 1, 4.94 ($SD = 3.83$) at Wave 2, and 4.81 ($SD = 3.86$) at Wave 3.

Risk attitudes were measured by asking participants how much they liked six types of risk behavior, that is, "smoking cigarettes," "riding a bicycle through red lights," "skipping school," "destroying or damaging things," "drinking alcohol," and "hitting someone." Answers could be given on a 5-point scale ranging from *I don't like to do this at all* (1) to *I like to do this very much* (5). The items measuring risk attitudes were sufficiently one-dimensional (as indicated by factor analysis), and the resulting scale was internally reliable (Cronbach's alpha varied between 0.75 and 0.86 for the three waves; see also Stark & Flache, 2012). Scores on these items were averaged and then rounded to the nearest integer to receive the original scale with 5 categories. Respondents scored on average 1.74 ($SD = 0.76$) on risk attitudes at Wave 1, 1.84 ($SD = 0.80$) at Wave 2, and 2.00 ($SD = 0.88$) at Wave 3.

Norm salience was based on the class-level correlations between status and risk attitudes at Wave 1. The status of each student was determined from peer nominations received for the question "Who do others want to be associated with ('Who is popular')?" In most studies of popularity among adolescents, respondents are asked to nominate the most (and least) popular peers; this can cover many aspects, such as influence, dominance, having social power, attractiveness, and resource control (cf. LaFontana & Cillessen, 2002; Lease et al., 2002; Parkhurst & Hopmeyer, 1998). In our measure, we added what adolescents presumably mean by saying that a person is popular, namely that people want to be connected with the popular person, to be associated with that person, and to "bask in reflected glory" (Dijkstra et al., 2010). This yielded a reputation-based measure for popularity, which was also used in other research showing similar features as the typical used question (Dijkstra et al., 2008, 2009).

Classes with a *neutral* norm salience were identified by a class-level correlation between status and risk attitudes below 0.10 (Type 1; $n = 19$). Classes with an insignificant but positive correlation between 0.10 and 0.30 were considered to have a *somewhat positive* norm salience (Type 2; $n = 17$). Finally, all classes with a significantly *positive* correlation (at the $p < .1$ level) above 0.30 were classified

as having a positive norm salience (Type 3; $n = 11$). Only the correlations at the first observation were relevant for this typology because we aimed to examine the contextual effects of norm salience in class on changes in adolescents' risk attitudes over time. Correlational analyses showed that risk attitudes were not more strongly correlated with status for boys and girls at Wave 1 in all three types of classes (descriptive statistics for correlations are available upon request). This means that norm salience did not differ for boys and girls in our data.

ANALYTIC STRATEGY

Analyses were conducted using longitudinal social network analysis. We applied the stochastic actor-based model of Snijders and colleagues (SIENA 4.0 in R), which allowed influence processes to be untangled from selection processes regarding risk attitudes (Snijders, Van de Bunt, & Steglich, 2010; Snijders et al., 2007). The effects were first analyzed for each class separately. To analyze the impact of norm salience on peer influence processes regarding risk attitudes, the results per class were then aggregated for the three types of classes (i.e., classes with a neutral norm salience, somewhat positive norm salience, and more positive norm salience) following Snijders and Baerveldt's meta-analysis method (2003). We also carried out these analyses at the school level and found similar patterns of results as at the classroom level; that is, in schools where status was positively associated with risk attitudes, peer influence was stronger (results available upon request). For a detailed explanation of the stochastic actor-based model, we refer to Snijders et al. (2007) and Veenstra, Dijkstra, Steglich, and Van Zalk (2013).

Network Effects

We included a number of parameters to control for structural effects of the network: outdegree (density), reciprocity, and transitivity (transitive triplets) (Veenstra & Steglich, 2012). *Outdegree* (density) represents the general tendency of network members (actors) to have friendship ties, measured by the number of friendship choices made in the network. *Reciprocity* represents the tendency to reciprocate friendship nominations in best friendship networks. *Transitive triplets* represent the tendency to nominate friends of someone's own friend, representing a closed group of three actors who are friends with each other.

Selection Dynamics

To examine selection effects based on risk attitudes and respondents' sex, we included several selection parameters. The *alter* effects indicate the effects of risk attitudes and sex on nominations received as best friend. Conversely, the *ego* effects are the effects of risk attitudes and sex on nominations given as best friend. Inclusion of both effects is necessary to provide more reliable estimates of the extent to which the adolescents formed new friendships with those who were similar in risk attitudes and sex. Selection of friends with similar risk attitudes was modeled by an interaction between the risk attitudes *alter* and *ego* effects. This *alter* $\times$ *ego* selection effect indicated whether adolescents with higher risk attitudes would choose as friends peers who had also higher risk attitudes. Similarity-based selection for sex was captured with the *same-sex* effect.

Influence Dynamics

Peer influence processes were investigated with the average *alter* parameter. This effect estimated whether adolescents whose friends had higher risk attitudes also developed higher risk attitudes themselves over time. The effect mirrors our measurement of similarity-based selection of friends with similarly high-risk attitudes in the influence part of the model. We considered peer influence to occur when adolescents adjusted their own level of risk attitudes to that of their friends who on average had higher risk attitudes. Peer influence processes were estimated while controlling for the linear shape parameter, quadratic shape parameter, and the main effect of sex on risk attitudes. Inclusion of these effects was necessary to provide more reliable estimates of the extent to which the adolescents' risk attitudes were subject to peer influence processes.

The *linear shape* parameter modeled the overall tendency toward risk attitudes. The *quadratic shape* effect modeled the feedback effect of the risk attitude on itself, which could be self-correcting (scores being pulled toward the average) or reinforcing (scores being pushed toward the extremes) depending on whether the quadratic shape effect was negative or positive, respectively. Finally, to control for possible sex differences in risk attitudes, we included the main effect of sex on risk attitudes (*Risk attitudes: effect of sex*), as it has been shown in previous studies that boys tend to be more deviant

than girls (De Cuyper et al., 2009; Knecht et al., 2010).

RESULTS

Descriptive Statistics

Table 1 provides information on the sample, network characteristics, and measures. Information on these aspects is provided separately for school classes with a neutral norm salience, somewhat positive norm salience, and more positive norm salience. In the first panel of Table 1, we see a nearly equal distribution of boys and girls in school classes with neutral and somewhat positive norm salience (around 50%), but a slightly higher distribution of boys in school classes with a more positive norm salience (around 60%).

In the second panel of Table 1, an initial description of the network structure is provided. Average outdegree indicates that the number of friendship nominations increased over time. The density indicates that respondents nominated around 20% of their classmates as friends over the three waves. The reciprocity parameter shows that respondents reciprocated more than half (50%–60%) of the friendship nominations they received from classmates by also nominating those classmates as friends. Transitivity is the ratio of the numbers of actually divided by potentially transitive triplets (50%–60%), reflecting the tendency of respondents to befriend the friends of friends. Respondents also had a strong tendency to nominate same-sex classmates as friends (80%–90%).

The fraction of stable friendship nominations relative to all new, lost, and stable friendships was around 40% (Jaccard Index), which made our data sufficiently stable for longitudinal social network analysis (Veenstra & Steglich, 2012). Overall, the different types of classes were similar with respect to the sociometric indices.

The third panel of Table 1 shows the development of risk attitudes over time. It appeared that the development of risk attitudes followed a similar pattern in all types of classes. Risk attitudes increased among all adolescents in all types of classes. When we look at the differences in risk attitudes between boys and girls, we see that girls scored on average lower on risk attitudes than boys. Differences in risk attitudes between boys and girls were significant at the separate measurement points, but marginal ($p < 0.10$) at the first and the third measurement point in classes with a more positive norm salience. No significant differences were found over time.

TABLE 1
Description of the Sample, Network Characteristics, and Measures for School Classes With a Neutral, Somewhat Positive, and More Positive Correlation Between Status and Risk Attitudes

Sample	Neutral classes			Somewhat positive classes			More positive classes		
	W 1	W 2	W 3	W 1	W 2	W 3	W 1	W 2	W 3
Respondents									
Present	408	403	385	367	374	364	231	222	227
Missing	18	26	51	24	19	28	14	24	23
Boys	53.2%	52.1%	52.2%	49.9%	50.5%	49.3%	58.4%	58.6%	58.6%
Friendship									
Number of ties	1536	2008	1843	1514	1793	1700	965	1131	1146
Average	3.77 (3.04)	4.98 (3.87)	4.79 (3.78)	4.13 (3.31)	4.79 (3.49)	4.68 (3.74)	4.18 (3.31)	5.09 (4.28)	5.05 (4.18)
outdegree									
Density	17.0%	21.4%	20.4%	17.7%	21.1%	21.1%	18.6%	21.9%	22.5%
Reciprocity	57.5%	60.7%	61.4%	62.0%	62.0%	59.1%	64.1%	61.3%	61.5%
Transitivity	56.3%	57.4%	57.4%	63.2%	60.5%	60.1%	57.7%	58.7%	60.0%
Same sex	83.7%	80.0%	78.5%	88.6%	88.2%	84.1%	87.4%	84.1%	82.2%
Risk attitudes									
Average boys	1.83 (0.77)	1.92 (0.79)	1.94 (0.83)	1.78 (0.77)	1.93 (0.77)	2.11 (0.92)	1.94 (0.84)	2.07 (0.94)	2.21 (0.97)
Average girls	1.61 (0.69)	1.72 (0.79)	1.85 (0.89)	1.59 (0.65)	1.64 (0.72)	1.82 (0.83)	1.75 (0.82)	1.78 (0.74)	1.98 (0.88)
Average popular	1.58 (0.58)	1.98 (0.79)	2.25 (0.93)	1.90 (0.62)	2.20 (0.78)	2.31 (1.03)	2.57 (0.84)	2.63 (0.83)	2.71 (1.06)
students									
Average	1.75 (0.76)	1.81 (0.80)	1.94 (0.83)	1.66 (0.73)	1.74 (0.74)	1.93 (0.86)	1.76 (0.75)	1.89 (0.86)	2.07 (0.93)
nonpopular									
students									
Network autocorrelation									
Moran's <i>I</i>	0.091	0.104	0.149	0.064	0.179	0.251	0.138	0.146	0.167
Sample change	W 1–W 2	W 2–W 3	W 1–W 3	W 1–W 2	W 2–W 3	W 1–W 3	W 1–W 2	W 2–W 3	W 1–W 3
Respondents									
Leaving	1	2	3	2	6	8	0	3	3
Joining	5	9	13	4	5	9	1	7	8
Staying	425	427	423	389	387	383	245	243	242
Friendship change									
Distance	1411	1289	1487	1259	1199	1363	743	784	849
Jaccard index	40.8%	45.9%	33.4%	42.1%	44.8%	34.5%	44.8%	44.6%	37.7%
Risk attitudes									
change									
Distance	152	164	200	151	153	176	94	107	133
Fraction	63.9%	59.2%	53.6%	59.8%	60.5%	54.1%	61.0%	57.1%	49.8%
stable									
actors									

Note. P 1 and P 2 refer to the period in between observations 1 and 2, and 2 and 3, respectively; standard deviations placed between brackets; density was calculated as N of ties divided by the total amount of ties; reciprocity was calculated as $2M/(2M + A)$, where M = mutual ties and A = asymmetric ties; transitivity was calculated as N of transitive triplets divided by N of 2-paths (or 2-stars).

Network Effects

The results of the SIENA meta-analysis are presented in separate models in Table 2. We first present the results of the analysis in which all 47 classes were included (Model 1). We then present the results for school classes with neutral, somewhat positive, and more positive norm salience (Models 2 through 4). Because the results of the network effects were similar in all models, we discuss the network effects based on Model 1.

The negative outdegree effect indicates the low density of the network and implies that friendships were not formed with just anyone in class. The positive reciprocity effect indicates that respondents reciprocated the friendship nominations they received from classmates. The positive transitive triplets effect reflects that respondents befriended the friends of their friends. Taken together, the network effects imply that participants had a tendency to keep the friendship networks closed, and this tendency toward network closure did not vary across classes with neutral, somewhat positive, or more positive norm salience (all $ps < .001$).

Selection Dynamics

Selection dynamics for sex were also similar across classes with a neutral, somewhat positive, and a more positive norm salience. The negative sex alter effects indicate that boys received on average fewer friendship nominations of classmates than girls. The positive sex ego effects indicate that boys gave on average more friendship nominations to classmates than girls. Finally, the positive same-sex effects show that boys and girls nominated on average more same-sex classmates as friends than cross-sex classmates.

For risk attitudes, none of the alter or ego effects reached conventional levels of significance. Apparently, respondents who scored higher on risk attitudes did not attract more best friendship nominations (alter effect) nor nominated more peers as best friend (ego effect). The selection similarity effect (i.e., risk attitudes alter $\times$ ego effect) was highly significant in all classes (Model 1). However, when the analyses were separated across classrooms with a neutral, somewhat positive, or more positive norm salience, we saw that the

TABLE 2

SIENA Meta-Analysis of Network and Influence Dynamics for Risk Attitudes in All Classes, and Classes With a Neutral, Somewhat Positive, and More Positive Correlation Between Status and Risk Attitudes

	All classes ^a Model 1		Neutral Model 2		Somewhat positive Model 3		More positive Model 4	
	Est.	SE	Est.	SE	Est.	SE	Est.	SE
Network effects								
Outdegree	-2.15***	.07 ^b	-2.07***	.11 ^b	-2.28***	.10 ^b	-2.08***	.14 ^b
Reciprocity	0.87***	.07 ^b	0.98***	.09 ^b	0.74***	.13 ^b	0.91***	.14 ^b
Transitive triplets	0.20***	.01	0.19***	.01	0.21***	.01	0.18***	.02
Selection dynamics								
Sex alter (1 = boy)	-0.20**	.06 ^b	-0.14	.08 ^b	-0.27*	.13 ^b	-0.25*	.10 ^b
Sex ego	0.21**	.07 ^b	0.14	.12 ^b	0.36*	.10 ^b	0.07	.11 ^b
Same sex	0.82***	.05 ^b	0.72***	.07 ^b	0.94***	.06	0.84***	.17
Risk attitudes alter	0.02	.03 ^b	-0.01	.07 ^b	0.06	.04	0.02	.05
Risk attitudes ego	-0.01	.03 ^b	-0.02	.05	-0.02	.06	0.01	.08
Risk attitudes alter $\times$ ego	0.20***	.04	0.19*	.08	0.33***	.07	0.11	.08
Influence dynamics								
Risk attitudes linear shape	-0.07	.05	-0.03	.06	-0.13	.12	-0.02	.06
Risk attitudes quadratic shape	-0.16***	.04	-0.20*	.08	-0.12	.08	-0.16	.08
Risk attitudes: effect of sex	0.07	.07	0.05	.11	0.22	.14	-0.10	.11
Risk attitudes average alter	0.49***	.11	0.31	.20	0.42*	.18	0.92***	.17
No. of classes	47		19		17		11	
No. of students	1092		439		400		253	

Note. ^aSeven classes did not converge; effect of time was included twice, and effect of sex was fixed once to reach convergence of the model; several parameters had to be excluded from the meta-analysis due to high standard errors (>11); from a total of 47 classes, 46 classes were included in the meta-analysis for the average alter parameter (used to measure peer influence), of which 19 classes had a neutral correlation between status and risk attitudes, 17 a somewhat positive correlation between status and risk attitudes, and 10 a more positive correlation between status and risk attitudes.

^bSignificant differences found between classes according to the Snijders and Baerveldt method (2003).

* $p < .05$; ** $p < .01$; *** $p < .001$ (p -values two-sided).

selection similarity effect was still significant in classes with a neutral norm salience (Model 2) and particularly in classes with a somewhat positive norm salience (Model 3), but was nonsignificant in classes with a more positive norm salience (Model 4). We tested for differences between parameter estimates with independent samples *t*-tests, assuming equal variance across groups (with estimates $\hat{\beta}_a$ and $\hat{\beta}_b$ and standard errors SE_a and SE_b , respectively), with the following formula $\hat{\beta}_a - \hat{\beta}_b / \sqrt{SE_a^2 + SE_b^2}$, which under the null hypothesis of equal parameters has an approximating standard normal distribution (see Ripley, Snijders, & Preciado, 2012). Differences in selection similarity between somewhat positive and more positive classes were significant, but differences between neutral and more positive classes were not ($t[27] = 2.11$, $p = .05$; $t[29] = 0.67$, $p = 0.51$, respectively). These results show that adolescents with higher risk attitudes selected each other more as friends in classes where there was a neutral or particularly somewhat positive association between status and risk attitudes.

Influence Dynamics

Students' risk attitudes changed in relatively similar ways across the different types of classes. First, the linear shape effects tell us that respondents had a low (but nonsignificant) tendency toward risk attitudes. In other words, the majority of the adolescents scored below the mean on the 5-point scale for risk attitudes. Second, the (marginally) significant negative quadratic shape effects indicate that over time respondents with higher risk attitudes were more likely to turn to lower values, whereas respondents with lower risk attitudes were more likely to turn to higher risk attitudes ("regression to the mean"). Finally, the nonsignificant main effects of sex on risk attitudes suggest that boys and girls did not differ in changes in risk attitudes over time. Although boys scored higher on risk attitudes than girls (see Table 1), boys and girls changed their risk attitudes in the same way after controlling for their initial attitudes at the first observation.

The peer influence effect (i.e., average alter parameter) in Model 1 was highly significant ($p < .001$). When the analyses were separated across classrooms with a neutral, somewhat positive, or more positive classes, we see, however, that the average alter parameter was nonsignificant in classes with a neutral norm salience (Model 2). But peer influence processes occurred in classes with a

somewhat positive norm salience (Model 3) and particularly in classes with a more positive norm salience (Model 4), indicating a steep increase in strength of peer influence processes as the association of status with risk attitudes increased. Differences in peer influence between neutral and more positive classes and between somewhat positive and more positive classes were also significant ($t[29] = 2.32$, $p = .03$; $t[27] = 2.03$, $p = .05$, respectively). These results are in line with our hypothesis that adolescents' risk attitudes proliferate more via peer influence processes in classes where there is a more positive association between status and risk attitudes at Wave 1 (OR [log odds] = 2.51) compared with classes with somewhat positive (OR = 1.52) or neutral association between status and risk attitudes at Wave 1 (OR = 1.36).

In addition, we tested whether our findings regarding peer influence processes were really the result of an injunctive norm salience effect and not the result of average attitudes of classmates (descriptive norms). To this end, we classified the classes into two categories according to the mean risk attitude of all students, using the mean score for risk attitudes at Wave 1 as cutoff point ($M = 1.74$, $SD = 0.76$). Results showed that peer influence was present in both groups ($n = 24$, high group: Est. = 0.40, $SE = 0.18$, $p = 0.036$; $n = 22$, low group: Est. = 0.57, $SE = 0.15$, $p = .001$), indicating that descriptive norms do not differentiate peer influence processes between classrooms.

Finally, we tested another alternative explanation. If popular students are more influential than others are, this might also have to do with them occupying positions in the network that allow them to exert more influence. If this is true, then peer influence on adolescents' risk attitudes might not be the result of contextual effects through injunctive norm salience, but merely the result of the opportunity, that is, having more friendship relations, to exert influence upon peers. This would mean that the correlation between popularity and best friendship nominations should be stronger in classes with a more positive norm salience, where we found stronger influence processes, than in classes with a somewhat positive or especially a neutral norm salience. To test this, we examined the correlations between popularity (i.e., number of status nominations received) and friendship (i.e., number of best friend nominations received) across the three types of classes. It appeared that in classes with a more positive norm salience, popular adolescents did not have more friendships, enabling them to be more influential, than in

classes with a somewhat positive or neutral norm salience (correlations available upon request). Instead, we found that popular adolescents were similar in number of best friendships across the three different types of school classes.

DISCUSSION

In this study, we examined the extent to which norm salience in school classes explain between-class variations in selection and particularly influence on adolescents' risk attitudes. The results showed that influence processes on adolescents' risk attitudes were more likely when there was a significantly positive association between status and risk attitudes in a class at the beginning of the school year. In these classes, adolescents seem to be more susceptible to peer influence. Our explanation is an injunctive norm salience effect: that is, high-status adolescents who also score high on risk attitudes set the norm in class. These adolescents are also more influential than their low-status peers. As a means of gaining status, low-status adolescents might conform to their high-status peers by adjusting their level of risk attitudes upwards. The alternative explanation, that is, descriptive norms reflected by the average attitudes of peers, did not differentiate peer influence processes between classrooms.

Another explanation is that popular adolescents are more influential because of having more friendship ties in classes with a more positive norm salience. In social network research, the position of actors matters for how others perceive them and are influenced by them (see Burt, 1976); especially, individuals who are more central in the network are in a better position (have more ties) to exert influence. The number of ties enables them to exchange more information to others in the network. Because popular students are also the ones who receive the most friendship ties (not controlling for everything else, for example, network structure, context; see above), they are likely also the ones who are most influential upon peers. Moreover, probably not all ties are "strong" (i.e., enduring reciprocal ties), but the amount of "weak" ties (i.e., dissolving unilateral ties) enables them to bridge the gap between peers, which makes them highly influential upon peers (see Granovetter, 1973). However, in this study, popular adolescents were similar in number of best friendships across the three different types of school classes, rejecting the possibility that popular students in classes with a more positive norm salience are

more influential because they have more opportunities to influence peers.

Although we did not *a priori* hypothesize about selection similarity effects, our results showed that selection effects for risk attitudes were not present in classes with a more positive norm salience, but were in classes with a neutral and particularly in classes with a somewhat positive norm salience. One explanation might be that in classes with a positive norm salience, adolescents with high-risk attitudes were attractive as friends to all classmates because of their popularity. The opposite might be true for students with high-risk attitudes in classes with no or less positive association between status and risk attitudes. In these classes, students with high-risk attitudes were not particularly popular, resulting in their seeking affiliation with similar peers. This fits the idea that risk-behaving, delinquent adolescents might be at the periphery of the peer group when located in contexts where risk behaviors are less rewarding (i.e., less or not associated with status), and hence more likely to seek each other's company (cf. Hirschi, 1969).

We theorized that low-status adolescents might conform to the behaviors or attitudes of high-status peers as a means of gaining status. In line with our expectation, adolescents whose friends had on average higher risk attitudes also developed higher risk attitudes themselves over time in classes with a more positive norm salience compared to classes with a somewhat positive norm salience or neutral norm salience. The positive estimates across neutral classes suggest that also in these classes risk attitudes increased among most adolescents (both the popular and the nonpopular ones; see results section; see also Table 1 for descriptive statistics). This increase, however, may not be "caused" by contextual effects through a norm salience effect, because the positive association between status and risk attitudes was absent in these classes. Instead, the increase might possibly reflect a natural development during adolescence when risk behaviors become more positively evaluated (Agnew, 2001; Allen, Weissberg, & Hawkins, 1989; Moffitt, 1993).

Building on this, an alternative explanation, which we could unfortunately not test, is that differences between classrooms reflect differences in maturation. It is possible that the individual development of high-status adolescents proceeds from being a "model child" (risk attitude average in neutral classes at Wave 1; see Table 1) to "rebellious adolescent" (risk attitudes average in more positive classes at Wave 3; see Table 1). More specifically, increases in risk attitudes in classes with a

more positive norm salience may reflect a more advanced maturation of the popular children in the class with nonpopular students following at a distance (cf. Moffitt, 1993). Future studies that look into status-based influence processes should also look into the explanation by mere maturation to rule out this explanation too.

In this study, we focused on peer influence processes in classes because students in the Netherlands spend most of their time in their own classroom. This results in more interaction and more status-based influence at the class level. However, evidence of peer influence on adolescents' risk behavior has also often been found at the grade level (De Cuyper et al., 2009; Mercken et al., 2009) and sometimes not or less at the classroom level (Knecht et al., 2010, 2011). This suggests that analyses at the class level may provide an incomplete picture because students naturally also have friends outside their own classroom. Grade-level networks may provide better insight into the actual influence processes among adolescents (Veenstra & Dijkstra, 2011). To this end, we also looked at the same processes on the school level, revealing that peer influence processes at the school level were similar to peer influence at the classroom level, particularly in contexts where we expected to find stronger influence processes (i.e., schools with a more positive norm salience). This indicates that in this sample the direction of influence processes was similar on class and school levels, indicating that the classroom is still a relevant context to study adolescents' development (Knecht et al., 2010; Stark & Flache, 2012). However, research has shown that out-of-school friends can be more influential than in-school friends when it comes to risk behaviors (e.g., Kerr, Stattin & Kiesner, 2007; Kiesner, Kerr, & Stattin, 2004). Unfortunately, our data did not allow testing the relative influence of out-of-school friends.

Furthermore, it should be kept in mind that the association between status and risk attitudes takes part in a broader context with school- and community-wide norms that are reinforced by the media. Although risk behavior emerges in adolescence, and is sometimes seen as normative (Moffitt, 1993), we still found interesting variation between classrooms, allowing us to understand why risk attitudes proliferate in some classes and not in others. More research on how norm salience reflect or contradict norms in specific contexts, such as schools or neighborhood, might be fruitful in furthering our understanding of the appeal of risk behavior in adolescence.

Limitations

A limitation is that we did not measure the influence of high-status adolescents on low-status adolescents directly; we measured it indirectly by capturing the contextual effect of norm salience (i.e., the positive association between status and risk attitudes) on peer influence processes. We provided a mean estimate of the peer influence processes in a particular classroom and, hence, did not test whether low-status adolescents were directly subject to the influence of their high-status peers. Complex interactions like these require high statistical power, which becomes problematic in a sample with small classes. However, influence processes are not necessarily bound to contact, but may also come about through observation. Less popular adolescents who are not befriended with their popular classmates may adopt their risk attitudes to become more popular. This suggests that popular adolescents are also influential when they are not directly connected with adolescents in their class.

Another limitation is that our focus on risk attitudes does not allow strong conclusions about actual risk behaviors. For the present study, we do not know to what extent risk attitudes will translate into actual behavior—or to what extent actual behavior earlier influenced the attitudes (Fishbein & Ajzen, 1975). Future research should look into the role of norm salience in actual behavior to gain more insights into the underlying processes resulting in risk behaviors and possible discrepancies between attitudes and actual behaviors.

CONCLUSION

The current findings show that the between-class variations in effects of influence and to a lesser extent selection processes on adolescents' risk attitudes can be explained by existing norm salience in the classroom. Our results imply that a simple aggregation of class-level results is insufficient to explain the dynamic nature of the selection and particularly influence processes that take place within these classroom networks. In support of our findings is the observation that classes that were allocated into three categories did not differ in their basic descriptive statistics or in the effects of our control variables in the SIENA model (i.e., network and sex effects). Classrooms were seemingly similar but differed in one respect: norm salience. This difference accounted for distinct influence as well as selection processes over the course of the

school year. The results show that adolescents were more susceptible to peer influence on risk attitudes over time in classes where risk attitudes were positively correlated with status at the start of the school year than in classes where risk attitudes were not or less correlated with status, suggesting status-based influence processes.

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